

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

Introduction

Low-contrast or poorly illuminated images often hide important visual details, making them difficult to interpret. Histogram-based enhancement techniques improve image visibility by modifying the distribution of pixel intensities.

In this experiment, two enhancement methods were applied to the same set of five degraded images:

- **Histogram Equalization (HE):** Histogram Equalization (HE) is an automatic contrast enhancement technique that redistributes the pixel intensity values of an image across the entire grayscale range (0–255). It calculates the cumulative distribution function (CDF) of the image histogram and remaps each pixel to a new intensity value based on this distribution. As a result, the contrast is increased, and details that were previously hidden in dark or bright regions become more visible.

First, the probability of occurrence of each intensity level is calculated as:

$$p(r_k) = \frac{n_k}{N}$$

where:

- n_k = number of pixels with intensity r_k
- N = total number of pixels

The cumulative distribution function (CDF) is:

$$CDF(r_k) = \sum_{j=0}^k p(r_j)$$

The new pixel intensity is obtained using:

$$s_k = (L - 1) \times CDF(r_k)$$

where:

- L = total number of gray levels (256 for an 8-bit image)
- s_k = transformed intensity value
- **Histogram Matching (HM):** also known as histogram specification, is a technique that modifies the histogram of an input image so that it resembles the histogram of a chosen reference image. Instead of simply spreading pixel values uniformly, it uses the cumulative distribution functions of both the input and reference images to map

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

pixel intensities accordingly. This produces a more controlled enhancement while preserving a natural appearance.

Methodology

The following steps were followed for each image:

1. Read the degraded image.
2. Convert it to grayscale.
3. Apply Histogram Equalization
4. Apply Histogram Matching using the first image as the reference.
5. Generate histograms for the original and processed images.
6. Compare the visual appearance and histogram distributions.

Code and Results

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image_paths = [
    "pic1.png",
    "pic2.png",
    "pic3.png",
    "pic4.png",
    "pic5.png"
]

def generate_histogram(image):
    histogram=cv2.calcHist(
        [image],
        [0],#for grey scale no plane,no rgb
```

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

None, #for mask whole image will be considered as 1 mask s we are ot divinding the image to get parts to do local operations

[256], #no of bins,no. of levels aas 8 bit imaage

[0,256]#[0,256)

)

return histogram

for input_path in image_paths:

input_image=cv2.imread(input_path)

gray_image=cv2.cvtColor(input_image,cv2.COLOR_BGR2GRAY)

input_hist=generate_histogram(gray_image)

intensity=np.arange(256)

plt.figure(figsize=(12,6))

plt.subplot(1,2,1)

plt.imshow(gray_image,cmap="gray")

plt.title("Input Image")

plt.axis("off")

plt.subplot(1, 2, 2)

plt.bar(intensity, input_hist)

plt.title("Input Image Histogram")

plt.xlabel("Intensity")

plt.ylabel("Frequency / number of pixels")

plt.xlim([0, 255])

plt.grid(alpha=0.25)

plt.show()

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

Histogram Normalisation

for input_paths in image_paths:

```
#input_image=cv2.imread(input_path)
#gray_image=cv2.cvtColor(input_image,cv2.COLOR_BGR2GRAY)
#input_hist=generate_histogram(gray_image)
ttl_pi=gray_image.size
normalized_hist=input_hist/ttl_pi
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plt.bar(intensity,input_hist,width=1)
plt.title("Original Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency / Number of Pixels")
plt.xlim([0, 255])
plt.grid(alpha=0.25)
# Normalized histogram
plt.subplot(1, 2, 2)
plt.bar(intensity, normalized_hist)
plt.title("Normalized Histogram")
plt.xlabel("Intensity")
plt.ylabel("Probability")
plt.xlim([0, 255])
plt.ylim([0, 1])
plt.grid(alpha=0.25)
```

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

plt.show()

Histogram Equalisation

```
def hist_equalisation(image):
```

```
    # gray_image=cv2.cvtColor(image,cv2.COLOR_BGR2GRAY)
```

```
    equalised_image=cv2.equalizeHist(gray_image)
```

```
    return equalised_image
```

```
for input_path in image_paths:
```

```
    input_image=cv2.imread(input_path)
```

```
    gray_image=cv2.cvtColor(input_image,cv2.COLOR_BGR2GRAY)
```

```
    equalized_image = hist_equalisation(input_image)
```

```
    original_hist=generate_histogram(gray_image)
```

```
    equalized_hist=generate_histogram(equalized_image)
```

```
    intensity=np.arange(256)
```

```
    plt.figure(figsize=(12,8))
```

```
    plt.subplot(2, 2, 1)
```

```
    plt.imshow(gray_image, cmap="gray")
```

```
    plt.title("Original Image")
```

```
    plt.axis("off")
```

```
    plt.subplot(2,2,2)
```

```
    plt.bar(intensity, original_hist)
```

```
    plt.title("Original Image Histogram")
```

```
    plt.xlabel("Intensity")
```

```
    plt.ylabel("Frequency")
```

```
    plt.xlim([0, 255])
```

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

```
plt.grid(alpha=0.25)
# 3. Equalized Image
plt.subplot(2, 2, 3)
plt.imshow(equalized_image, cmap="gray")
plt.title("Equalized Image")
plt.axis("off")
# 4. Equalized Histogram
plt.subplot(2, 2, 4)
plt.bar(intensity, equalized_hist)
plt.title("Equalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 255])
plt.grid(alpha=0.25)
plt.tight_layout()
plt.show()
```

Histogram matching

```
from skimage.exposure import match_histograms
for input_pth in image_paths:
    input_image = cv2.imread(input_pth)

    # Convert the image to grayscale
    gray_image = cv2.cvtColor(input_image, cv2.COLOR_BGR2GRAY)
```

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

Load the reference image

```
reference_path = "image.png"
```

```
reference_image = cv2.imread(reference_path)
```

Convert the reference image to grayscale

```
gray_reference = cv2.cvtColor(reference_image, cv2.COLOR_BGR2GRAY)
```

```
matched_image = match_histograms(gray_image, gray_reference)
```

```
matched_image.dtype
```

Convert the matched image to 8-bit format

```
matched_image = np.uint8(matched_image)
```

Generate the histogram of the original image

```
original_hist = generate_histogram(gray_image)
```

Generate the histogram of the reference image

```
reference_hist = generate_histogram(gray_reference)
```

Generate the histogram of the matched image

```
matched_hist = generate_histogram(matched_image)
```

Create intensity values from 0 to 255

```
intensity = np.arange(256)
```

Display images and histograms

```
plt.figure(figsize=(15, 9))
```

Display the original image

```
plt.subplot(2, 3, 1)
```

```
plt.imshow(gray_image, cmap="gray")
```

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

```
plt.title("Original Image")
```

```
plt.axis("off")
```

```
# Display the reference image
```

```
plt.subplot(2, 3, 2)
```

```
plt.imshow(gray_reference, cmap="gray")
```

```
plt.title("Reference Image")
```

```
plt.axis("off")
```

```
# Display the matched image
```

```
plt.subplot(2, 3, 3)
```

```
plt.imshow(matched_image, cmap="gray")
```

```
plt.title("Histogram Matched Image")
```

```
plt.axis("off")
```

```
# Display the original histogram
```

```
plt.subplot(2, 3, 4)
```

```
plt.bar(intensity, original_hist, width=1)
```

```
plt.title("Original Image Histogram")
```

```
plt.xlabel("Intensity")
```

```
plt.ylabel("Frequency / number of pixels")
```

```
plt.xlim([0, 255])
```

```
plt.grid(alpha=0.25)
```

```
# Display the reference histogram
```

```
plt.subplot(2, 3, 5)
```

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

```
plt.bar(intensity, reference_hist, width=1)
plt.title("Reference Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency / number of pixels")
plt.xlim([0, 255])
plt.grid(alpha=0.25)
```

```
# Display the matched histogram
plt.subplot(2, 3, 6)
plt.bar(intensity, matched_hist, width=1)
plt.title("Matched Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency / number of pixels")
plt.xlim([0, 255])
plt.grid(alpha=0.25)
```

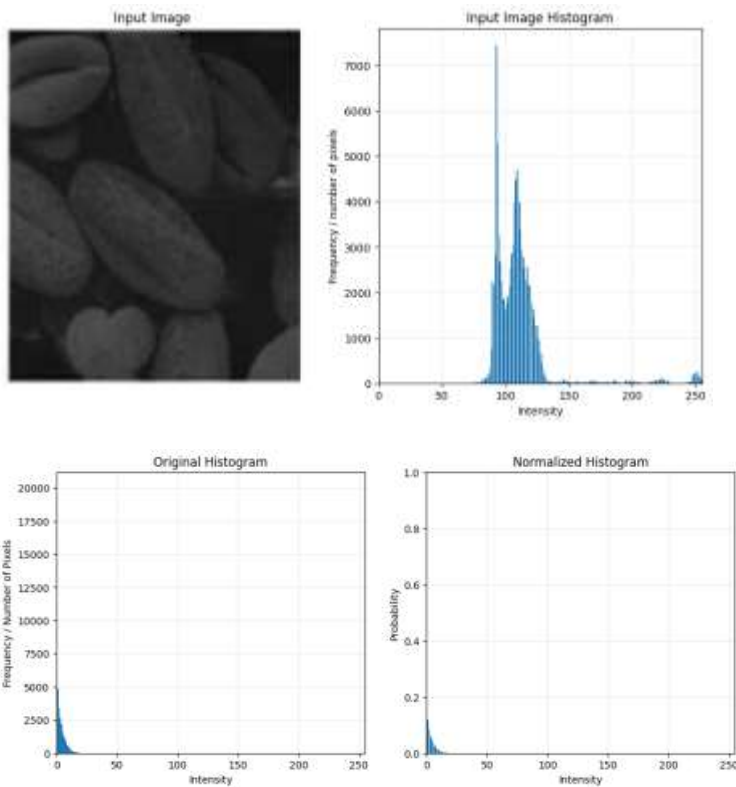
```
# Adjust spacing
plt.tight_layout()
```

```
# Display the results
plt.show()
```

Results and Observations

Image 1

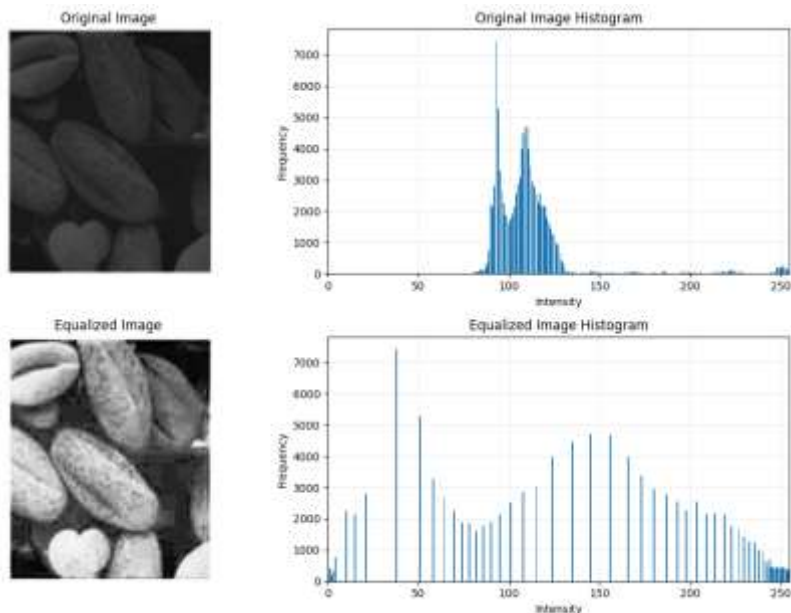
 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037



Histogram Equalization

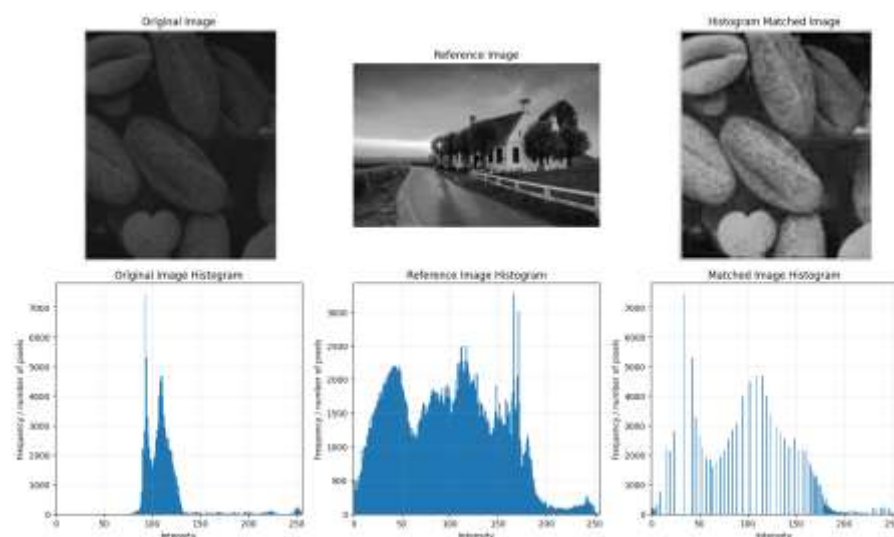
- Increased the overall brightness and contrast.
- The texture on individual seeds became more visible.
- The histogram spread across a wider intensity range.

 Marwadi University Marwadi Chandanina Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037



Histogram Matching

- Produced a smoother enhancement.
- Avoided excessive brightness.

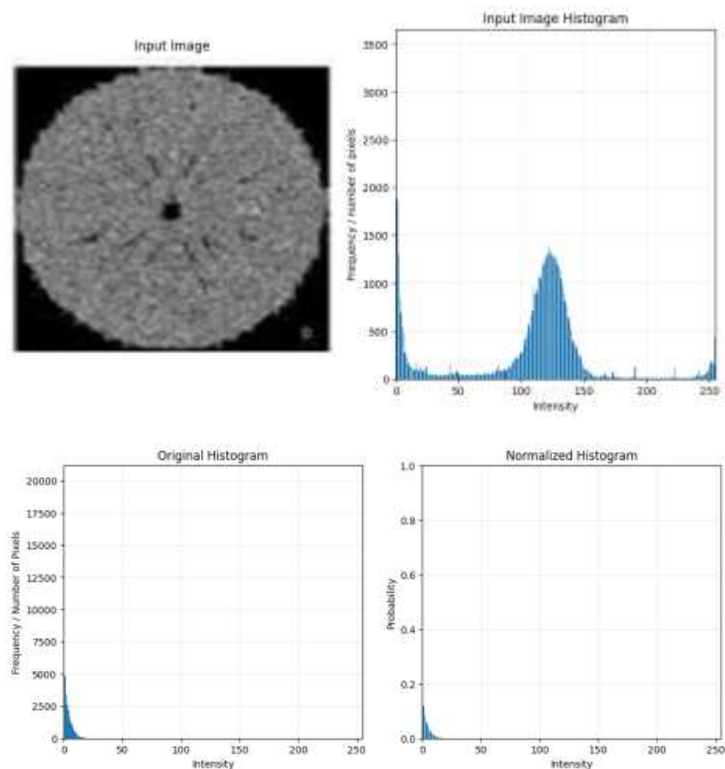


Better Method: Histogram Equalization

Reason: The seed textures became sharper and more distinguishable.

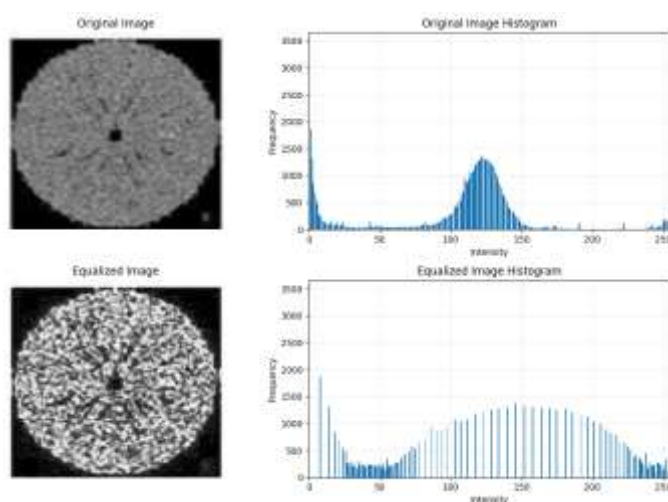
 Marwadi University Marwadi Chandanina Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

Image 2



Histogram Equalization

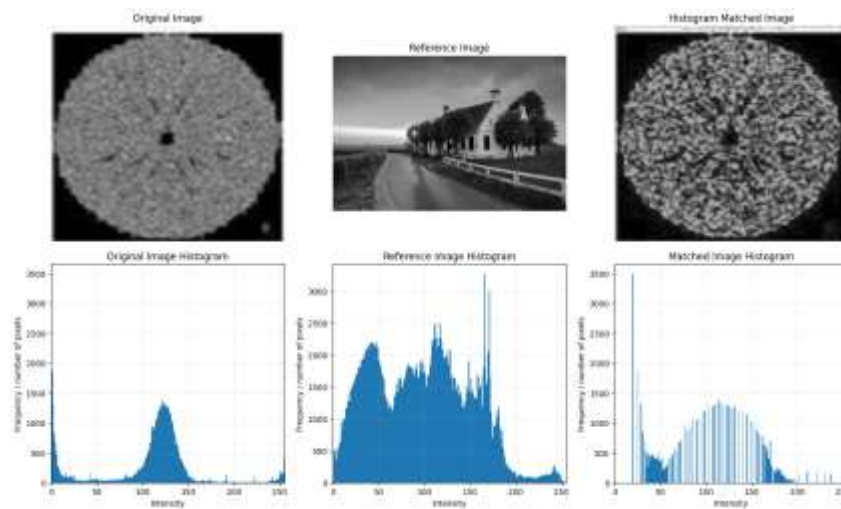
- Enhanced internal structures significantly.
- Made edges and fine patterns easier to observe.
- Increased global contrast.



 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

Histogram Matching

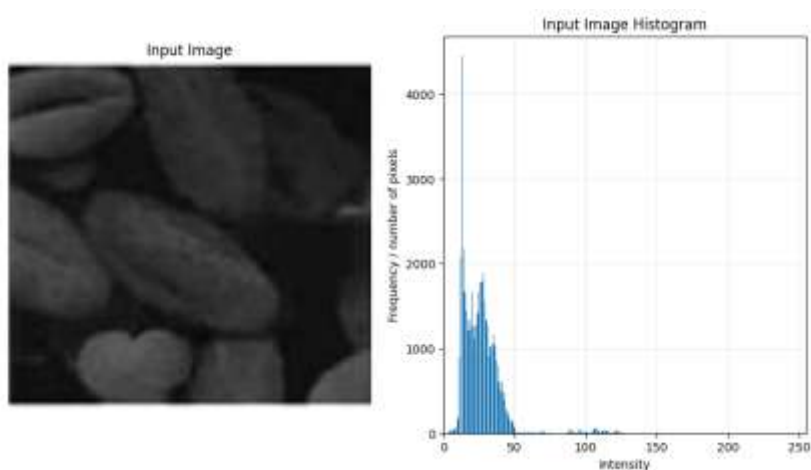
- Improved clarity without making the image overly bright.
- Produced a softer appearance.



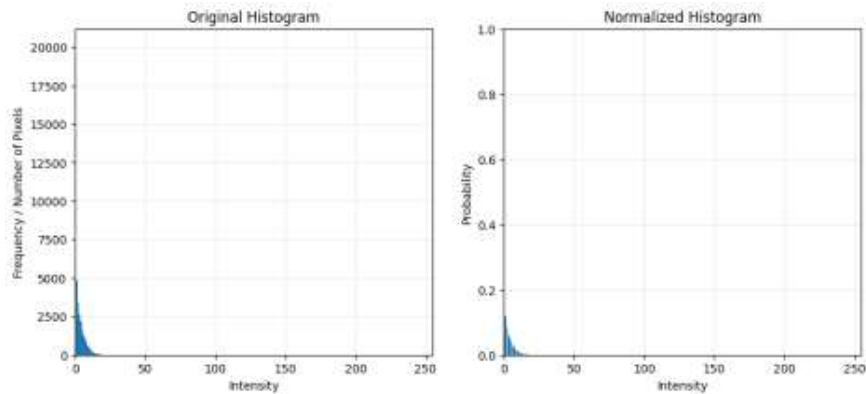
Better Method: Histogram Equalization

Reason: More internal details became visible.

Image 3

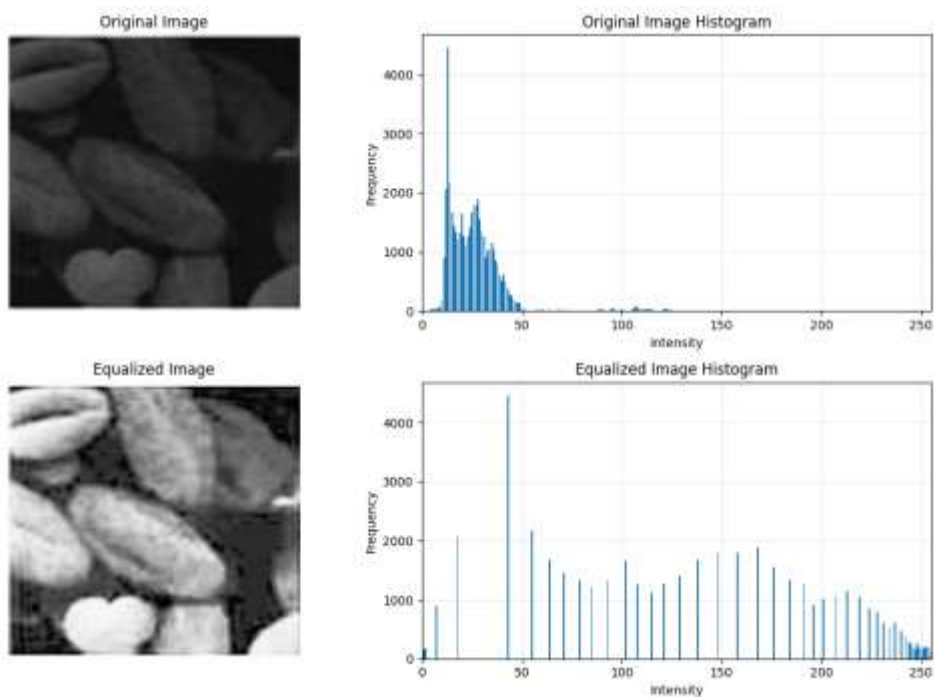


 Marwadi University Marwadi Chandanina Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037



Histogram Equalization

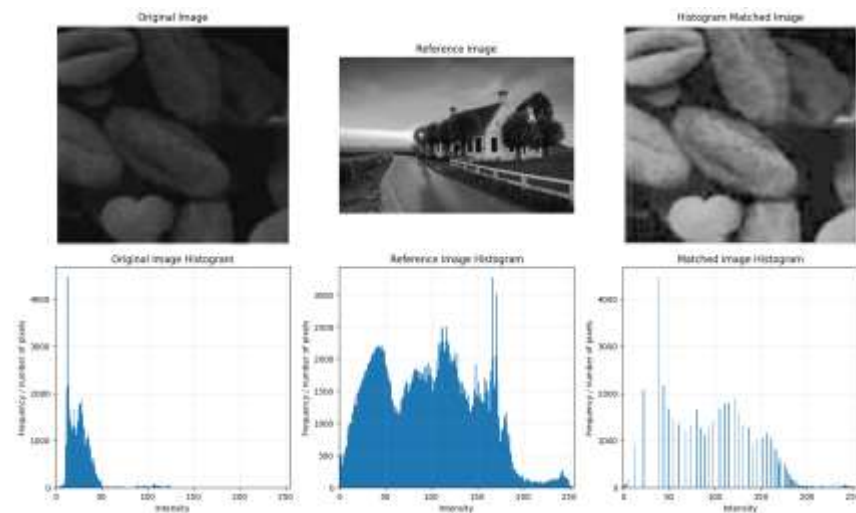
- Brightened the image considerably.
- Revealed previously hidden seed outlines.
- Slightly increased noise.



Histogram Matching

- Improved visibility while keeping brightness balanced.
- Preserved smoother intensity transitions.

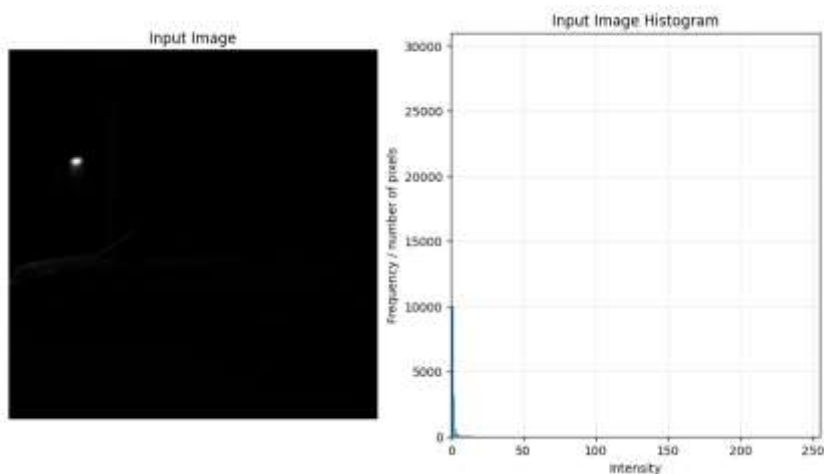
 Marwadi University Marwadi Chandanina Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037



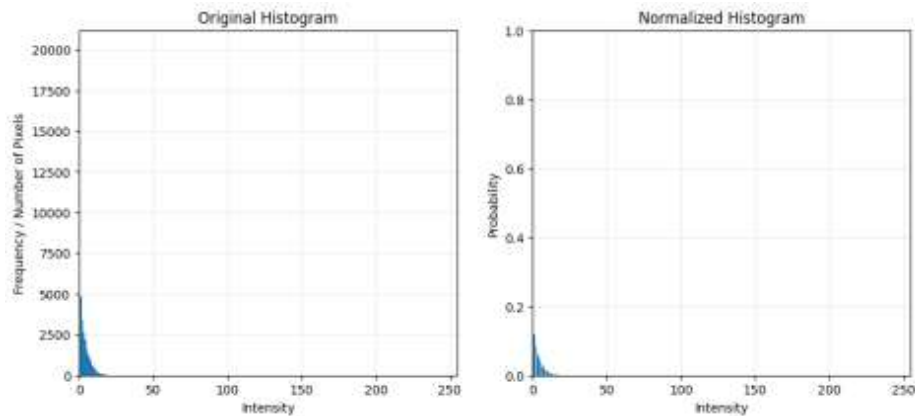
Better Method: Histogram Matching

Reason: It provided better visibility without over-enhancing the image.

Image 4

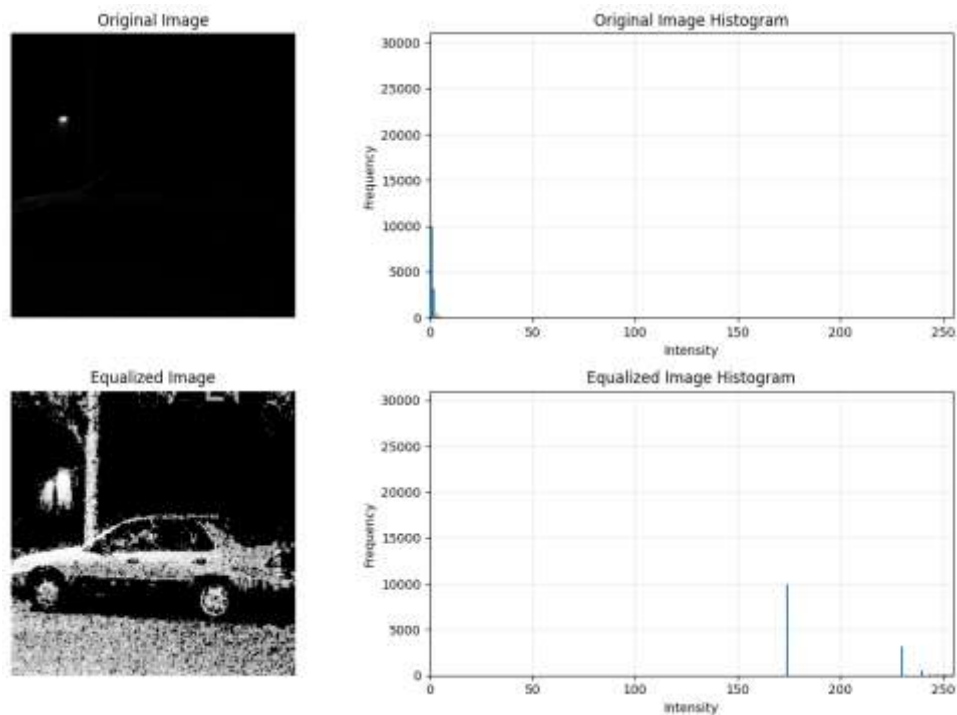


 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037



Histogram Equalization

- Revealed the car and road details.
- Increased noise in the dark regions.
- Produced a harsher appearance.

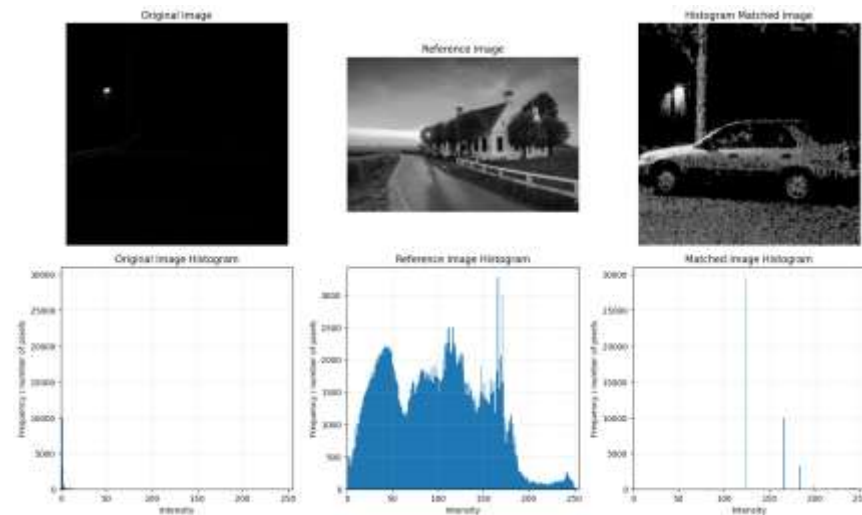


Histogram Matching

- Preserved the nighttime atmosphere.
- Improved visibility with less noise amplification.

 Marwadi University Marwadi Chandanina Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

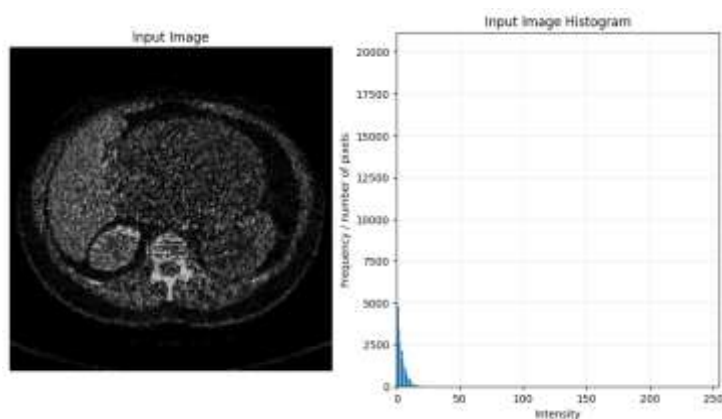
- Produced a more balanced result.



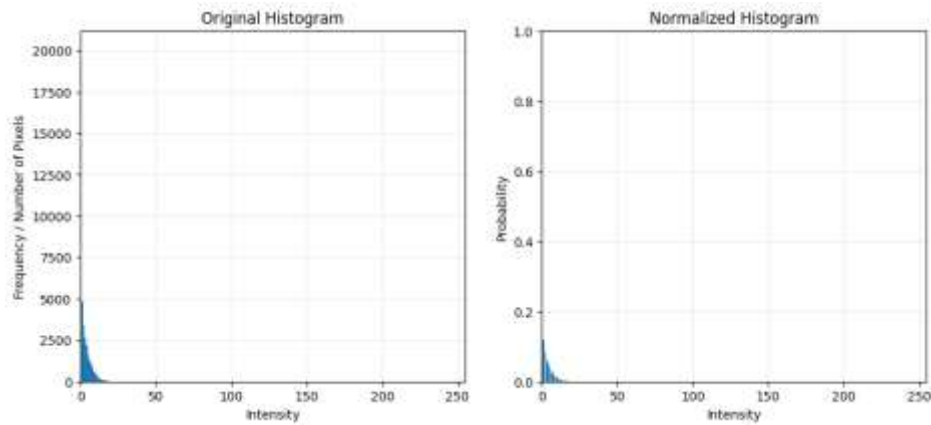
Better Method: Histogram Matching

Reason: The enhancement looked more natural.

Image 5

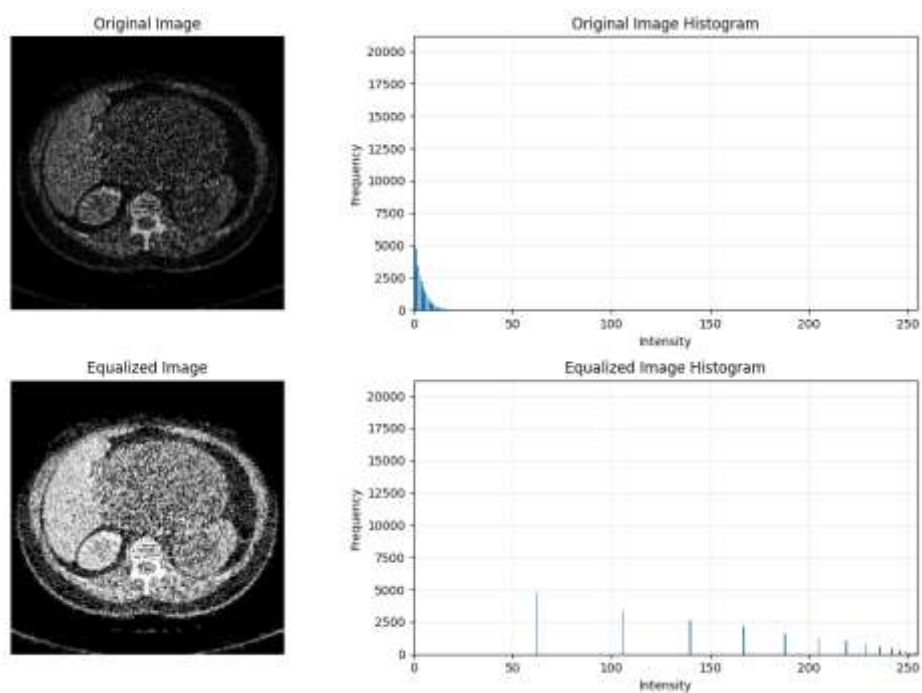


 Marwadi University Marwadi Chandanina Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037



Histogram Equalization

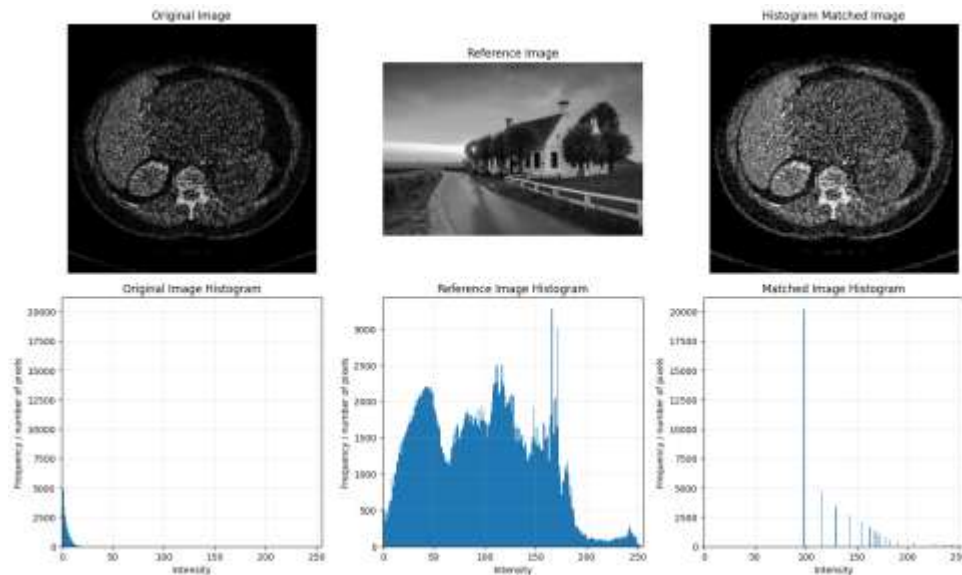
- Increased the visibility of anatomical boundaries.
- Introduced noticeable graininess.



Histogram Matching

- Preserved realistic grayscale intensities.
- Enhanced important structures while reducing harshness.

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037



Better Method: Histogram Matching

Reason: Medical images benefit from preserving natural intensity relationships.

How Each Technique Changes Contrast and Clarity

Histogram Equalization

- Redistributes pixel values across the full intensity range.
- Increases global contrast.
- Makes hidden details more visible.
- Can also amplify noise in dark regions.
- Works best for images with uniformly low contrast.

Histogram Matching

- Adjusts the histogram to resemble a reference image.
- Produces controlled and natural-looking enhancement.
- Preserves brightness relationships better.
- Avoids excessive contrast enhancement.
- Depends on selecting a suitable reference image.

 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: apply histogram equalization and histogram matching techniques to enhance their visual quality.	
Open-ended : 8	Date: 23/08/2026	Enrollment No: 92400133037

Histogram Behaviour

The histogram changes clearly illustrate how both methods operate.

- Original images: Pixel intensities were concentrated within a narrow range, indicating low contrast.
- Histogram Equalization: The histograms became more evenly distributed across the grayscale range, increasing overall contrast.
- Histogram Matching: The histograms shifted toward the reference distribution, producing smoother intensity transitions while preserving a more natural appearance.

Conclusion

In this experiment, both Histogram Equalization and Histogram Matching successfully improved the visual quality of degraded images, but they performed differently depending on the image characteristics.